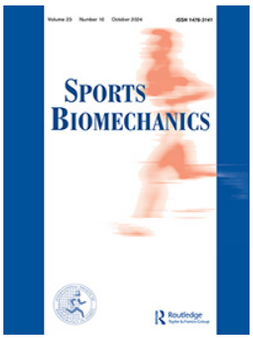

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Spatiotemporal kinematics during top speed sprinting in male intercollegiate track and field and team sport athletes

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ABSTRACT

We investigated spatiotemporal kinematics during top speed sprinting and biomechanical running strategies in 98 male intercollegiate athletes from a range of athletic backgrounds in track and field (TF, $n = 28$) and team sports (TS, $n = 70$). Participants completed 40 m running trials with sagittal plane motion analyses of high-speed video captured from 30 m to 40 m. Across the entire sample, measures of contact time, step rate, step length, flight length and duty factor (ratio of contact duration to stride duration) were meaningfully correlated with top speed ($p < 0.05$, $0.51 \leq |r \text{ or } \rho| \leq 0.78$). Flight time and contact length were weakly correlated with top speed ($p < 0.05$, $0.27 \leq |r \text{ or } \rho| \leq 0.34$). When comparing sub-groups of Slow TF ($n = 14$) and Fast TS athletes ($n = 22$) with similar top speeds (~9.3 m/s), Fast TS athletes clearly demonstrated a more ground-based strategy, with longer ground contact times and contact lengths, shorter flight times and flight lengths, and larger duty factors. Therefore, the results of this study suggest that existing technical models and normative metrics based on data from TF athletes could require modification when evaluating and coaching sprinting performance with TS athletes.

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KEYWORDS
Running biomechanics; top speed sprinting; locomotor performance; track and field; team sports

Introduction

Linear sprinting speed is crucial for athletes in both track and field and team sports. In track and field, sprinting speed is the essential determinant of performance in the short sprint events (Haugen et al., 2018; Slawinski et al., 2017) and a crucial factor in field events such as the horizontal jumps (Linthorne, 2008). In team sports, linear sprinting speed can play a large role in on-field outcomes (Faude et al., 2012) and talent selection (Robbins, 2010). Although acceleration during the initial stages of a sprint is obviously key (Morin et al., 2015; Rabita et al., 2015), the ability to attain a fast top speed is also very important for both track and field (TF) and team sport (TS) athletes (Clark et al., 2019; Slawinski et al., 2017).

Logically, a large body of research has examined the determinants of top speed sprinting performance. With respect to spatiotemporal kinematics, faster top speeds are typically associated with shorter ground contact times, more frequent step rates and longer step

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lengths (Clark & Weyand, 2014; Douglas et al., 2020; Nummela et al., 2007; Weyand et al., 2000). Duty factor, the ratio of contact duration to total stride duration, typically decreases with faster running speeds (Hanley et al., 2022). Other spatiotemporal variables, such as flight time and contact length (horizontal distance travelled by the body during the ground contact phase), are often relatively similar even for runners of different top speeds (Weyand et al., 2000, 2010). Flight length (horizontal distance travelled by the body during the flight phase [Hunter et al., 2004]) should be related to top speed (Paradisis et al., 2019) given that flight times and contact lengths usually fall within a narrow range.

Building on some of the foundational performance variables listed above, other analyses have developed approaches for characterising running styles and kinematic strategies. Tools have been established to assess gait as more ground-based ('terrestrial') or flight-based ('aerial') when evaluating upright running style at submaximal speeds (Gindre et al., 2016; Hanley et al., 2022; Lussiana et al., 2017; Patoz et al., 2019). In this classification scheme, terrestrial runners have a style that includes increased leg compression during ground contact and decreased vertical displacement during flight, whereas aerial runners are characterised by a bouncing gait with decreased leg compression during ground contact and increased vertical displacement during flight (Gindre et al., 2016).

Furthermore, although sprinting technique for TF versus TS athletes is a popular discussion point for researchers and practitioners (Clark & Jeffreys, 2024), to date there have been very few studies to directly compare the top speed sprinting mechanics of these groups. Specifically, previous investigations have examined TF versus TS athletes at their maximum sprinting performance (Clark & Weyand, 2014; Douglas et al., 2020), with TF athletes exhibiting faster top speeds compared to TS. Another important study has explored the mechanical differences between TF and TS athletes at similar sub-maximal sprinting speeds (Colyer et al., 2018). In all three of these studies, there have been different ratios of contact and flight time for TF versus TS athletes, with TS athletes demonstrating a contact duration that is a greater percentage of total stride duration compared to TF athletes. However, at present, research comparing TF and TS of similar top speed capabilities is limited and warrants further investigation.

Therefore, in this study we investigated the determinants of top speed in male intercollegiate athletes from a range of sporting backgrounds (TF and TS athletes). Participants from these backgrounds were chosen to represent athletes who sprint regularly as part of their sport (TS) versus athletes who are trained to sprint as their sport (TF). Our first hypothesis was that across the entire sample, contact time, step rate, step length, flight length and duty factor would be meaningfully correlated with top speed, but that flight time and contact length would not. Our second hypothesis was that when analysing TF and TS athletes of comparable top speeds, TS athletes would attain similar speeds utilising a more ground-based strategy, with relatively longer ground contact times and contact lengths, relatively shorter flight times and flight lengths, and greater duty factors.

Materials and methods

Participants

A total of 98 male varsity intercollegiate student-athletes volunteered for the study, with data collected at three universities. All participants provided written informed

consent in accordance with their university's respective Institutional Review Boards, which had approved the study (West Chester University: IRB# FY2022-361; University of Pennsylvania: IRB# 852005; Princeton University: IRB# 15345). The study was conducted in accordance with the Declaration of Helsinki. In alignment with the inclusionary criteria, all participants were at least 18 years of age, injury free and actively practicing with their team during the testing period, and had passed a medical exam before joining their team for the season. Participants included 28 track and field athletes (TF) and 70 team sport athletes (TS). All TF athletes competed in the sprints (≤400 m), horizontal jumps and/or hurdle events. All TS athletes competed in lacrosse ($n = 41$), soccer ($n = 18$) or baseball ($n = 11$). Complete participant descriptive characteristics are displayed in Table 1.

Testing procedures

Testing was performed on a standard rubberised running track at each university. An indoor track was used at one university and outdoor tracks were used at the other two universities. Data collection was performed on the outdoor tracks when the weather was temperate, the track was dry, and there were no substantial headwinds or tailwinds.

Upon obtaining signed consent forms, the height and weight of each participant was measured. Participants then completed their usual full-body warm-up routine as prescribed by their coaches, which included activities such as jogging, skipping, dynamic stretches and sub-maximal sprints. The entire testing and data collection process was completed within a single session, lasting less than 90 min. Participants completed all physical activities while wearing their own athletic clothing and running sneakers (track spikes were not permitted for TF athletes).

For the experimental testing, participants completed two 40 m running trials with video and timing data recorded during each trial. Participants began each trial in an upright, two-point stance, with their preferred leg positioned forward, and began running at their own initiative. Participants were instructed to perform an all-out acceleration from the beginning of the sprint and maintain maximum speed until they crossed the finish line at the 40 m mark. A minimum of 4 min of rest between trials was provided to allow for complete recovery.

Data collection and analysis

Sagittal plane video was recorded with an Apple iPhone 14 Pro Max (Apple USA, Cupertino, CA). The iPhone main camera (48 megapixels) was used in slow motion mode (240 frames/s) with all other controls at default settings. The field of view captured

Table 1. Descriptive statistics for all participants and for the participants divided into groups of Track and Field (TF) and Team Sport (TS) athletes. All values are listed as mean ± standard deviation.

	Age (years)	Height (m)	Mass (kg)	Body Mass Index (kg/m^2)
All Participants ($n = 98$)	19.6 ± 1.3	1.80 ± 0.06	81.2 ± 8.8	24.9 ± 2.4
All TF ($n = 28$)	18.8 ± 1.2	1.79 ± 0.05	75.1 ± 6.1	23.6 ± 2.1
All TS ($n = 70$)	19.9 ± 1.3	1.81 ± 0.07	83.6 ± 8.6	25.5 ± 2.3

by the camera was 10 m in length, from 30 m to 40 m in the running lane. The camera was mounted to a tripod at a height of 1.0 m, placed approximately 10 m perpendicular to the running lane and centred at 35 m. The 10 m field of view ensured that a complete stride cycle (two complete steps) could be captured. An automatic timing system (Speed Trap II, Brower Timing Systems, Draper, UT) was used along with video-based metrics. The split time from 30 m to 40 m was measured using this timing system, and for each participant the sprint trial with the faster recorded 30–40 m split time was used for the video analysis. A total of 98 videos were analysed, the fastest trial for each of the 28 TF and 70 TS athletes.

The video files were uploaded to Kinovea (v0.9.5) for 2D motion analysis and digitising. Kinovea is free software which has been collaboratively developed worldwide under the GPL v2 licence. Spatial calibration was performed by manually digitising each corner of the 30–40 m field of view: near lane line at 30 m, far lane line at 30 m, near lane line at 40 m and far lane line at 40 m. The stride cycle occurring in the most central field of view near 35 m was analysed and consisted of five video frames: first touchdown, first takeoff, second touchdown, second takeoff and third touchdown. The frames of touchdown and takeoff were defined as the first and last frames where the running shoe was clearly in contact with the ground. Foot position markers were then manually digitised for the heel and toe at each of the five frames of the stride cycle. The heel was digitised at the most inferior and posterior point on the shoe and the toe was digitised at the most inferior and anterior point on the shoe. All videos were analysed by the same investigator (first author). To evaluate intra-rater reliability, a sub-set of ten videos were digitised two times with a minimum of several weeks between analyses (see Results).

The manually digitised position data files were exported from Kinovea and imported into a custom Microsoft Excel spreadsheet to calculate the experimental variables under investigation in this study. Raw data were calibrated using the spatial calibration to create a pixels-per-metre conversion factor. This conversion factor was used to transform the positions of each marker to a coordinate system with units of metres. The foot-segment positions were determined from the heel and toe markers, and leg lengths L_0 were estimated from the body height measurements ($L_0 = 0.53 \bullet$ height) (Winter, 2009). The values for the experimental variables of interest were then calculated at each video frame of analysis.

Experimental variables

All reported values were trial averages from one complete stride cycle in the most central field of view consisting of one step each of right and left limbs.

Contact time

Contact time t_c (s) was determined from the time interval from touchdown to takeoff of the same foot.

Flight time

Flight time t_f (s) was determined from the time interval from takeoff of one foot to touchdown of the contralateral foot.

Step rate

Step rate *SR* (steps/s) was determined from the step time t_{step} (s) interval:

$$t_{step} = t_c + t_f \tag{1}$$

$$SR = \frac{1}{t_{step}} \tag{2}$$

Step length

Step length *SL* (m) was determined from the horizontal distance between the foot-segment midpoints from touchdown of one foot to touchdown of the contralateral foot.

Running speed

Running *Speed* (m/s) was determined from the average step length and average step rate during the two steps analysed in the central field of view:

$$Speed = SL \bullet SR \tag{3}$$

Contact length

Normalised contact length L_c/L_0 (dimensionless) was determined from the running speed, contact time, and leg length L_0 (m). It provides a measure of the horizontal distance travelled by the body during ground contact L_c (m) relative to the leg length L_0:

$$\frac{L_c}{L_0} = \frac{Speed \bullet t_c}{L_0} \tag{4}$$

Flight length

Normalised flight length L_f/L_0 (dimensionless) was determined from the running speed, flight time, and leg length L_0 (m). It provides a measure of the horizontal distance travelled by the body during the flight phase L_f (m) relative to the leg length L_0:

$$\frac{L_f}{L_0} = \frac{Speed \bullet t_f}{L_0} \tag{5}$$

Duty factor

Duty factor *DF* (dimensionless) was determined from the contact time and stride time ($2 \bullet t_{step}$). It provides the proportion of the stride when the foot is in contact with the ground:

$$DF = \frac{t_c}{2 \bullet t_{step}} \tag{6}$$

Statistical analysis

To test the first hypothesis, correlational analysis was performed across the entire sample of participants to determine the relationship between top speed and the following seven

dependent variables: contact time, flight time, step rate, step length, contact length, flight length and duty factor. For each variable, the normality of data was checked using the Shapiro-Wilk test. Normally distributed data were analysed with Pearson's r and non-normally distributed data were analysed with Spearman's ρ. Correlation coefficients were interpreted as follows ($|r$ or $\rho|$): 0.00–0.19 (very weak), 0.20–0.39 (weak), 0.40 to 0.59 (moderate), 0.60–0.79 (strong) and 0.80–1.00 (very strong) (Campbell, 2021). Only dependent variables that were significant and had correlation coefficients that were moderate or larger ($0.40 \leq |r$ or $\rho|$) were deemed meaningfully correlated with top speed. For graphic purposes, simple linear regression was also performed to generate a best-fit equation with variable x representing top speed. For the regression analyses, data were checked for normality by visually evaluating the residuals with Q-Q plots.

To facilitate interpretation of the data, participants were categorised into five sub-groups: Fast TF (top speed >9.65 m/s [$n = 14$]), Slow TF (top speed <9.65 m/s [$n = 14$]), Fast TS (top speed >9.00 m/s [$n = 22$]), Intermediate TS (top speed 8.50 to 9.00 m/s [$n = 29$]) and Slow TS (top speed <8.50 m/s [$n = 19$]). For the sprint variables and anthropometric variables, mean ± standard deviation (SD) values were calculated for each sub-group.

To test the second hypothesis, sub-groups of 'Slow TF' and 'Fast TS' athletes were evaluated to examine differences in running mechanics for athletes with comparable top speeds. This included 14 Slow TF athletes and 22 Fast TS athletes with similar mean top speed for each sub-group (Slow TF: 9.34 ± 0.21 m/s; Fast TS: 9.31 ± 0.19 m/s). After confirming the normality of data using the Shapiro-Wilk test, separate independent t-tests were conducted for each of the seven dependent variables to specifically examine significant differences between the sub-groups of Slow TF and Fast TS athletes. Absolute and percentage differences between the two sub-groups were also quantified, with percentage difference computed as: $(|Slow\ TF - Fast\ TS|)/((Slow\ TF + Fast\ TS)/2) \bullet 100$.

For all statistical tests, the *a priori* threshold for significance was set at $\alpha = 0.05$. The Bonferroni correction was used to adjust for multiple comparisons (Campbell, 2021), with an adjusted $\alpha = 0.007$. All statistics were completed using Microsoft Excel and GraphPad Prism software (version 9, San Diego, CA).

Results

The TF group had an average top speed of 9.67 ± 0.39 m/s (mean ± SD), with a range of 8.96 to 10.35 m/s. The TS group had an average top speed of 8.79 ± 0.46 m/s, with a range of 7.61 to 9.70 m/s. For the intra-rater reliability tests for the first and second intra-rater analyses, the mean percentage difference for each variable were as follows: top speed (0.34%), contact time (1.63%), flight time (1.47%), step rate (0.20%), step length (0.17%), contact length (1.54%), flight length (1.62%) and duty factor (1.61%).

As it relates to the first hypothesis, Figure 1 displays data for all participants across the range of top speeds, including the following variables: (1a) contact time, (1b) flight time, (1c) step rate, (1d) step length, (1e) contact length, (1f) flight length and (1 g) duty factor. For each of these variables, the best-fit linear regression equations, correlation coefficients and p-values are listed in the Figure 1 panels. Five of the dependent variables were meaningfully correlated with top speed, including three variables (contact time, flight length and duty factor) that demonstrated strong correlations with $0.69 \leq |r$ or $\rho| \leq 0.78$,

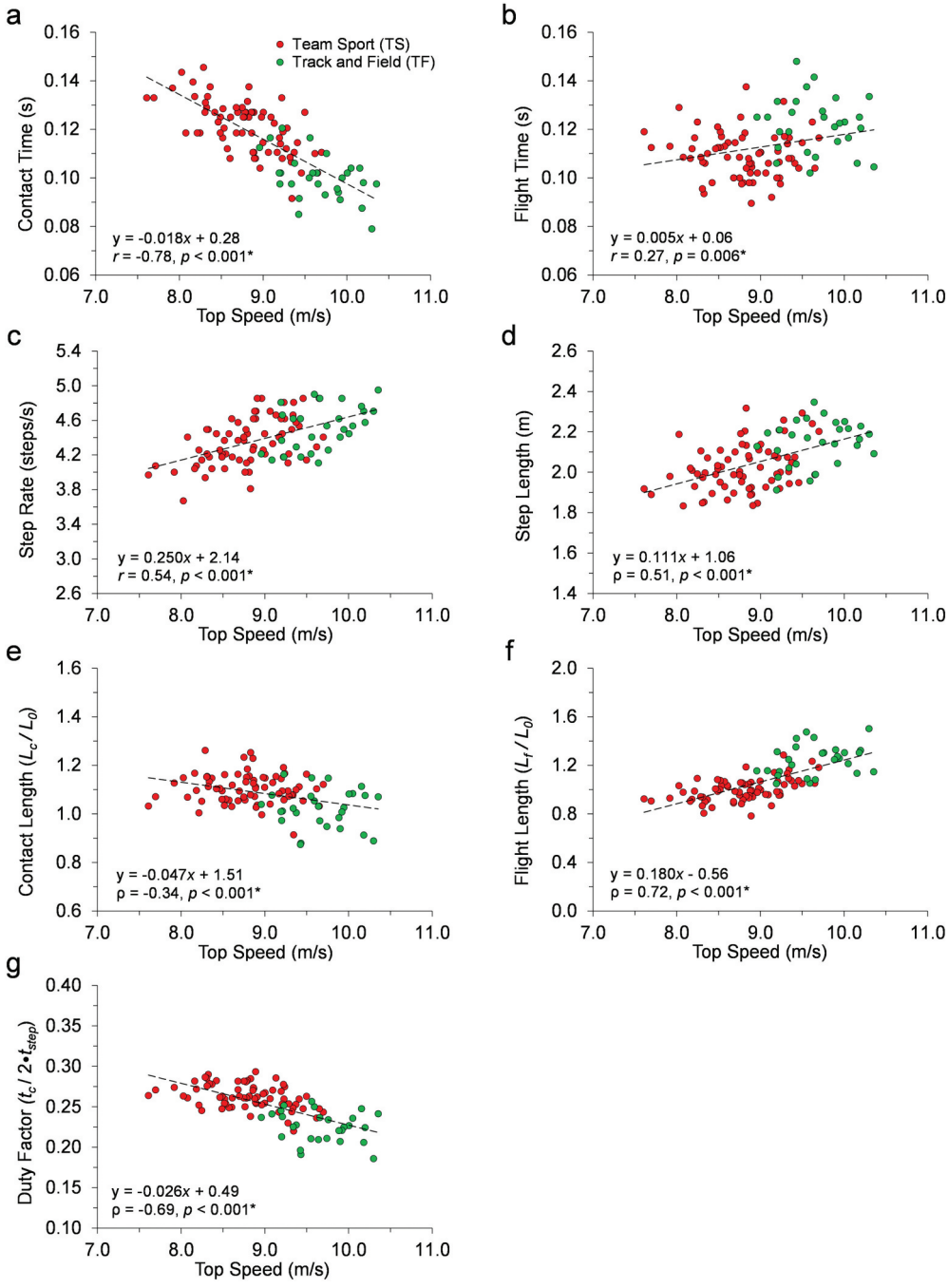

Figure 1. Data for all participants across the range of top speeds, including: (a) contact time, (b) flight time, (c) step rate, (d) step length, (e) contact length, (f) flight length, and (g) duty factor. Listed in the panel for each variable are the best-fit linear regression equations, correlation coefficients (Pearson's *r* or Spearman's ρ) and *p*-values (* indicates significant).

and two variables (step rate and step length) that demonstrated moderate correlations with $0.51 \leq |r \text{ or } \rho| \leq 0.54$. Two of the dependent variables (flight time and contact length) were significant but not meaningfully correlated with top speed, demonstrating weak correlations with $0.27 \leq |r \text{ or } \rho| \leq 0.34$. Additionally, for descriptive purposes and to facilitate interpretation of the data, Table 2 presents means and standard deviations with the participants categorised into the five sub-groups of Fast TF, Slow TF, Fast TS, Intermediate TS and Slow TS.

With respect to the second hypothesis, there were no significant differences between the Slow TF and Fast TS for top speed ($\Delta = 0.03$ m/s, 0.3%, $p = 0.636$). Furthermore, as illustrated in Figure 2, although these two sub-groups did not demonstrate significant differences for step rate or step length (Figure 2(c,d)), significant differences did exist between Slow TF and Fast TS for all other dependent variables. Specifically, Fast TS athletes achieved similar top speed compared to Slow TF but did so with a more ground-based strategy including longer ground contact times (Figure 2(a)) and contact lengths (Figure 2(e)), shorter flight times (Figure 2(b)) and flight lengths (Figure 2(f)), and greater duty factors (Figure 2(g)). Additionally, for the anthropometric variables, absolute and percentage differences between Slow TF and Fast TS were as follows: height ($\Delta = 0.04$ m, 2.1%), body mass ($\Delta = 7.2$ kg, 9.2%) and body mass index ($\Delta = 1.1$ kg/m^2, 4.7%).

Discussion and implications

In agreement with the first hypothesis, shorter ground contact times, more frequent step rates, longer step lengths and longer flight lengths, and smaller duty factors were all meaningfully related to top speed across the entire sample. These findings were expected when analysing all participants in this study and are in alignment with other investigations examining the spatiotemporal kinematics of top speed sprinting across a heterogeneous group of runners (Clark et al., 2020; Nummela et al., 2007; Weyand et al., 2000) or when comparing athletic groups of different sprinting abilities (Bushnell & Hunter, 2007; Douglas et al., 2020; Kunz & Kaufmann, 1981; Paradisis

Table 2. Descriptive statistics for participants divided into five sub-groups based on athletic background (track and field [TF] or team sport [TS]) and top speed: fast TF, slow TF, fast TS, intermediate TS and slow TS. All values are listed as mean ± standard deviation.

	Fast TF ($n = 14$)	Slow TF ($n = 14$)	Fast TS ($n = 22$)	Intermediate TS ($n = 29$)	Slow TS ($n = 19$)
Top Speed (m/s)	10.00 ± 0.21	9.34 ± 0.21	9.31 ± 0.19	8.78 ± 0.14	8.21 ± 0.25
Contact Time t_c (s)	0.097 ± 0.008	0.104 ± 0.010	0.114 ± 0.010	0.121 ± 0.009	0.130 ± 0.008
Flight time t_f (s)	0.120 ± 0.009	0.124 ± 0.013	0.109 ± 0.009	0.108 ± 0.010	0.111 ± 0.009
Step rate *SR* (steps/s)	4.62 ± 0.20	4.41 ± 0.27	4.50 ± 0.24	4.38 ± 0.27	4.16 ± 0.21
Step length *SL* (m)	2.17 ± 0.08	2.13 ± 0.13	2.08 ± 0.12	2.01 ± 0.12	1.98 ± 0.09
Contact Length L_c (m)	0.97 ± 0.07	0.97 ± 0.09	1.06 ± 0.08	1.06 ± 0.07	1.07 ± 0.07
Normalised Contact Length (L_c/L_0)	1.02 ± 0.08	1.03 ± 0.08	1.10 ± 0.06	1.11 ± 0.07	1.11 ± 0.06
Flight Length L_f (m)	1.20 ± 0.09	1.16 ± 0.13	1.02 ± 0.09	0.95 ± 0.08	0.91 ± 0.07
Normalised Flight Length (L_f/L_0)	1.27 ± 0.10	1.23 ± 0.15	1.06 ± 0.10	0.99 ± 0.08	0.95 ± 0.07
Duty Factor ($t_c/2 \bullet t_{step}$)	0.223 ± 0.016	0.228 ± 0.021	0.255 ± 0.016	0.264 ± 0.013	0.270 ± 0.013
Height (m)	1.79 ± 0.05	1.78 ± 0.06	1.82 ± 0.07	1.81 ± 0.07	1.81 ± 0.05
Body Mass (kg)	75.8 ± 5.7	74.5 ± 6.7	81.7 ± 8.3	82.4 ± 7.5	87.5 ± 9.6
Body Mass Index (kg/m^2)	23.6 ± 2.0	23.6 ± 2.3	24.7 ± 1.9	25.3 ± 1.8	26.6 ± 2.9

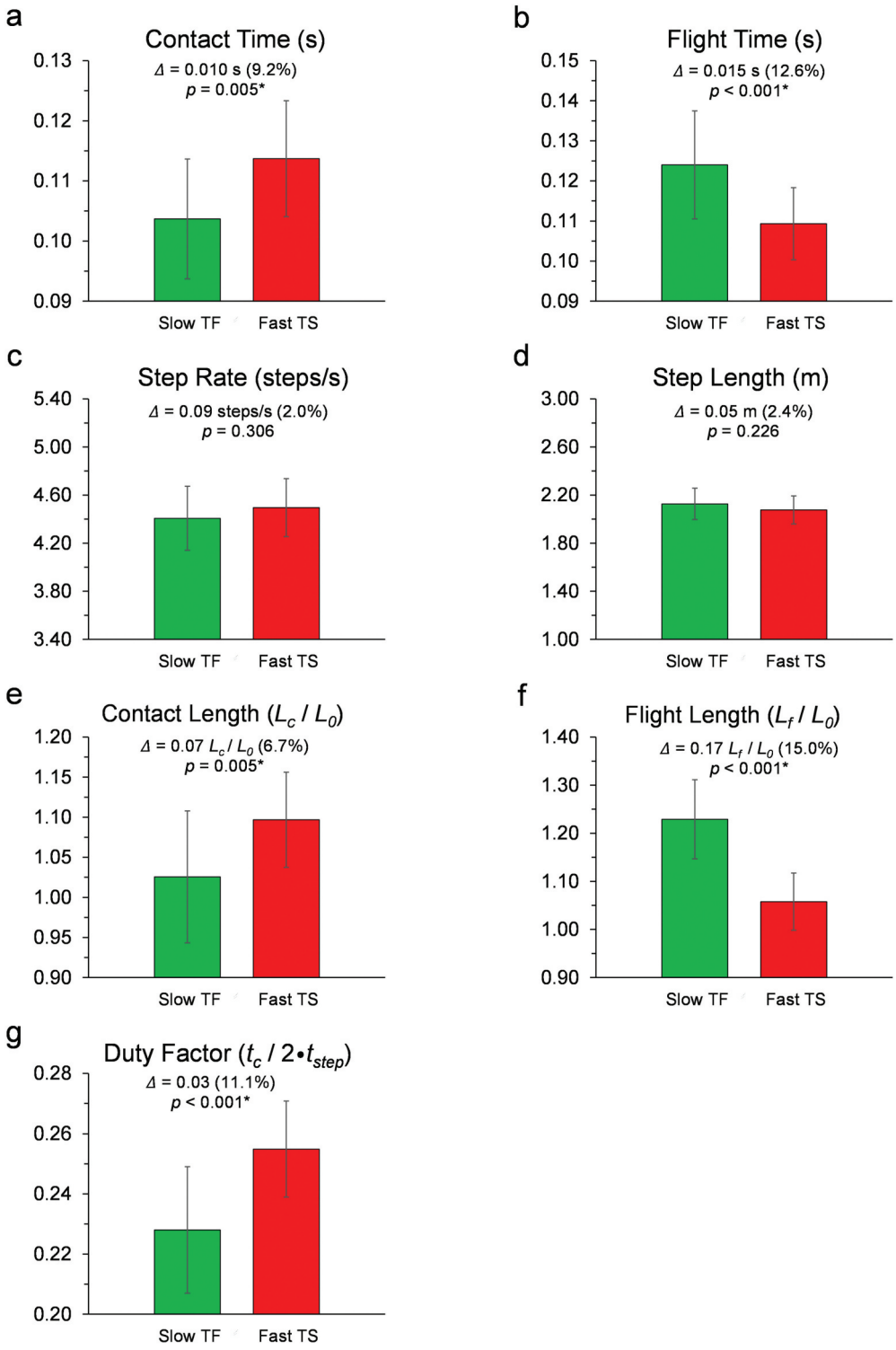

Figure 2. Results of the independent *t*-tests comparing slow track and field (TF) versus fast team sport (TS) for all sprint variables, including: (a) contact time, (b) flight time, (c) step rate, (d) step length, (e) contact length, (f) flight length, and (g) duty factor. Listed in the panel for each variable are absolute and percentage differences between slow TF and Fast TS, and the accompanying *p*-values from the independent *t*-tests (* indicates significant).

et al., 2019). The variables of flight time and contact length were significant but not meaningfully correlated with top speed across the entire sample (Figure 1(b,e)), and this agrees with prior research that has not demonstrated these variables to be primary determinants of sprinting performance (Clark & Weyand, 2014; Clark et al., 2020; Weyand et al., 2000).

With respect to the second hypothesis, the findings from the present study agree with the limited prior research comparing TF and TS at similar fast running speeds. Specifically, different ratios of contact time to flight time have been observed in TF athletes compared to TS athletes when running at different maximal speeds (Clark & Weyand, 2014; Douglas et al., 2020) or at similar sub-maximal speeds (Colyer et al., 2018). However, the present investigation highlights these different running strategies in TF versus TS athletes while sprinting at comparable top speeds. One worthwhile consideration is the speeds achieved by the Fast TS sub-group, which had a top speed range of approximately 9.0–9.7 m/s. Although this speed range represented the fastest ~ 30% of all TS athletes in the current study, it is unknown at present whether TS athletes with an even faster top speed (e.g., greater than 10.0 m/s) would demonstrate a similar ground-based strategy.

Particularly intriguing were the findings for duty factor, both as it related to correlations with top speed for the entire population (first hypothesis) and to comparisons of running strategies for Slow TF versus Fast TS (second hypothesis). When running at top speed, prior research has highlighted that faster runners typically apply larger vertical forces during shorter ground contact times (Clark & Weyand, 2014; Weyand et al., 2000) and have greater measures of leg stiffness (Paradisis et al., 2019) compared to slower runners. There is a link between the spatiotemporal kinematics and the kinetics, as decreased duty factors have been associated with increased measures of both vertical force and leg stiffness (Hanley et al., 2022; Patozet al., 2020). For the first hypothesis in the present study, it is therefore logical to suggest that the inverse correlation between duty factor and top speed across the entire sample (Figure 1(g)) indicates greater vertical force and leg stiffness for faster runners. For the second hypothesis, it is noteworthy that Fast TS had a substantially greater duty factor than Slow TF and yet were able to attain the same mean group speed. This would indicate that the ground-based strategy demonstrated by Fast TS potentially allowed them to attain the same top speed as Slow TF but with lower measures of vertical force and leg stiffness.

Results such as the above for the second hypothesis demonstrate the need for movement categorisation, which aligns with recent trends in sport science that have placed special emphasis on athlete profiling (Morin & Samozino, 2016). This has become specifically evident for the analysis of running mechanics, with the development of approaches for characterising running styles and kinematic strategies (Gindre et al., 2016; Hanley et al., 2022; Lussiana et al., 2017; Wild et al., 2022). Based on the results of this study, one intriguing question that arises is whether it is logical to expect that faster TS athletes will 'look' exactly like TF athletes during top speed sprinting. In fact, it is possible that the ground-based mechanics demonstrated by the Fast TS sub-group in the present data indicate that there could be some benefit for TS athletes to adopt this strategy when playing their sport, perhaps related to the change-of-direction and agility requirements of field- or court-based team competition (Clark & Jeffreys, 2024). The reactive nature of team sport competition might favour a slightly lower position of the

body and limbs, whereas track and field athletes typically execute pre-planned race models that allow for more upright positions in the top speed phase.

Additionally, although there are well-established technical models and normative data for elite sprinters in TF (e.g., Mann & Murphy, 2022), whether these exact models should apply to TS athletes is less clear at present. It is certainly possible for top speed mechanics to be modified, and specific sprint training interventions that guide TS athletes closer to a TF technical model might correspond with increases in top speed (e.g., Mendiguchia et al., 2021). However, while there are sure to be some similarities in body postures and segment positions shared by TF athletes and faster TS athletes, the more ground-based strategy exhibited by the Fast TS sub-group suggests that existing technical models and normative metrics derived from TF athletes could require modification when coaching TS athletes specialising in field- or court-based team sports.

Another interesting consideration of the ground-based strategy demonstrated by the TS athletes is the effect on energetics. Specifically, classic locomotor research suggests a precise linkage between ground contact times, rate of force application, and metabolic rate (Kram & Taylor, 1990). Therefore, the briefer ground contact times generally observed with faster top sprinting speeds might come at the expense of greater energy expenditure. Although performance in singular short sprint efforts is not limited by energy supply (Bundle & Weyand, 2012), additional considerations clearly exist with the repeated sprints that are fundamental components of game play in team sports. Therefore, it is possible that the ground-based strategy demonstrated by Fast TS athletes might strike an optimum blend enabling these athletes to attain relatively fast top speeds but doing so with a kinematic strategy that allows them to do so with longer ground contact times. Presumably, this could be advantageous if it reduces energy expenditure in a repeated-sprint event such as a team-based sporting competition.

From an experimental design and data collection perspective, a few limitations existed in this study. A field of view had to be established to capture top speed video for both the TF and TS sample of athletes. Prior research (Matsuo et al., 2019) has indicated that during experimental data collection, both team sport athletes and trained track and field athletes typically achieve maximum velocity, or an extremely high percentage of maximum velocity, by approximately 35 m into a sprint. Hence, to capture the top speed of the diverse group of participants involved in this study, a standardised field of view covering 30–40 metres was used. At this point in the sprint, upright running mechanics have been achieved, and the net vertical and horizontal accelerations occurring during one stride cycle were considered negligible. The discrete-frame video analyses of two steps limited the possibilities of performing investigations of inter-limb asymmetry or coordination. Additionally, since the participants of this study were all males, had comparable body dimensions, and wore normal athletic sneakers while sprinting on standard rubberised running tracks, the influences of anthropometric factors and foot-ground running interface were not studied.

This investigation specifically analysed spatiotemporal kinematic variables and the standardisations within the experimental protocol reduced the number of confounding variables and enabled efficient testing of a large sample of athletes with consistent results in the evaluation of intra-rater reliability. Future research should incorporate female athletes for similar studies, as well as analyse joint angle and body position kinematics for TF and TS athletes during top speed sprinting. This would lend physical context to the

spatiotemporal whole-body kinematic variables measured in this study leading to more appropriate technical models and coaching metrics for evaluation and training of TF versus TS athletes.

Summary and conclusion

In this study we investigated the spatiotemporal determinants of top speed running in male intercollegiate athletes from a range of sporting backgrounds in track and field (TF) and team sports (TS). When examined across the entire sample of athletes, measures of contact time, step rate, step length, flight length, and duty factor were meaningfully correlated with top speed. Flight time and contact length were weakly correlated with top speed. Although longer step lengths and more frequent step rates were associated with top speed across the entire sample, there were no significant differences in these two variables when comparing sub-groups of Slow TF and Fast TS athletes with similar top speeds (~9.3 m/s). However, compared to the Slow TF, the Fast TS athletes had longer ground contact times and contact lengths, shorter flight times and flight lengths, and greater duty factors. This indicates that the Fast TS athletes used a more ground-based running strategy, possibly optimising for the more reactive agility demands and repetitive sprint requirements of team sport competition. Thus, development of more specific technical models and normative metrics might be beneficial for coaching and for maximising sprinting performance with TS athletes.

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